# Computer Vision Practical Report:

# MOVE IMAGE FROM ONE PLACE TO ANOTHER

## 1. AIM

To read an input image using OpenCV, perform image translation (moving the image to a new location) using a transformation matrix, display the original and translated images side-by-side using Matplotlib, and save the translated image to disk.

## 2. PROCEDURE

- Load the input image and get its width and height.
- Define shifting distances along the X and Y axes ($t_x$ and $t_y$).
- Create a $2 \times 3$ translation matrix using NumPy.
- Shift the image location using `cv2.warpAffine()`.
- Display the original and shifted images side-by-side using Matplotlib.
- Save the output image to disk using `cv2.imwrite()`.

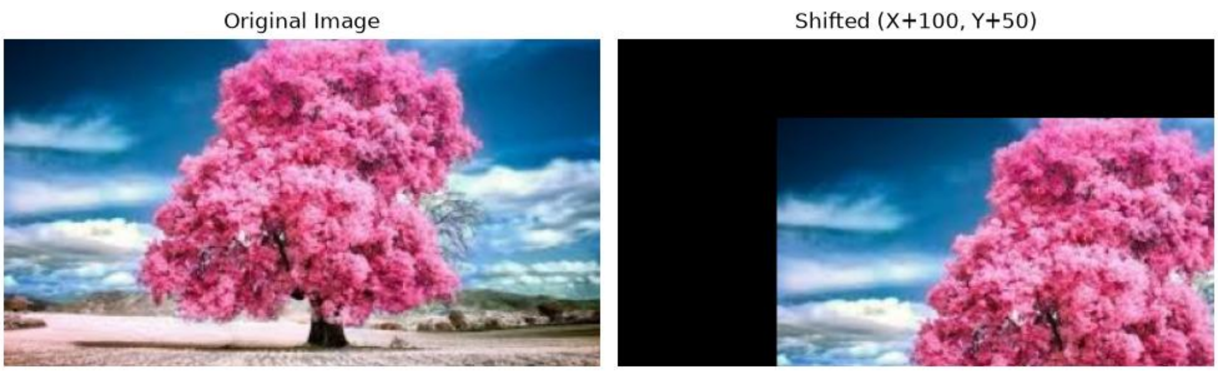
## 3. PROGRAM CODE

```python
import os
import cv2
import matplotlib.pyplot as plt

folder_path = r"C:\Users\adminuser\Desktop\Python CV"

image_files = [f for f in os.listdir(folder_path) if f.lower().endswith(('.png', '.jpg', '.jpeg', '.bmp'))]

if not image_files:
    print(f"No image files found in {folder_path}!")
    print("Files in folder:", os.listdir(folder_path))
    exit()

image_path = os.path.join(folder_path, image_files[0])
```

```python
print(f"Loading image: {image_path}")

img_bgr = cv2.imread(image_path)

if img_bgr is None:
    print(f"Error: Could not load image from {image_path}")
    exit()

img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

height, width, channels = img_rgb.shape

tx, ty = 100, 50

translated_img = [[[0, 0, 0] for _ in range(width)] for _ in range(height)]

for y in range(height):
    for x in range(width):
        new_x = x + tx
        new_y = y + ty
        if 0 <= new_x < width and 0 <= new_y < height:
            translated_img[new_y][new_x] = img_rgb[y][x]

plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(translated_img)
plt.title(f'Shifted (X+{tx}, Y+{ty})')
plt.axis('off')
```

```
plt.tight_layout()
plt.show()
```

## 4. OUTPUT




## 5. RESULT

The Python script executed successfully. The input image was loaded.